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23EAC301
Database Management Systems
Lab Sheet – 2

Topics : SQL Query (create table, insert into, select)

Exercise 1.1: Create a table called employee with the following columns:

Column	Type	Constraint
emp_id	INT	Primary Key, Auto Increment
name	VARCHAR(100)	Not Null
email	VARCHAR(100)	Not Null, Unique
department	VARCHAR(50)	—
salary	DECIMAL(10,2)	Must be > 0
hire_date	DATE	—
is_active	BOOLEAN	Default TRUE

Exercise 1.2: insert the following data to employee table:

name	email	department	salary	hire_date
Alice Roy	alice@company.com	Engineering	75000	2020-06-15
Bob Sharma	bob@company.com	Marketing	55000	2019-03-01
Carol Das	carol@company.com	Engineering	82000	2021-09-10
David Nair	david@company.com	HR	48000	2022-01-20
Eva Mehta	eva@company.com	Marketing	61000	2018-11-05
Farhan Khan	farhan@company.com	Engineering	91000	2017-07-14
Gauri Iyer	gauri@company.com	HR	52000	2023-02-28
Hari Pillai	hari@company.com	Engineering	70000	2020-12-01

Exercise 1.3: Write queries for the following (Basic SELECT Queries)

1. Display all details of employees.
2. Show only name and salary for all employees.
3. Display the names of employees working in the "HR" department.
4. Display employees whose department is not "HR".
5. Display employees whose department is either 'IT' or 'Finance'.
6. Show employees with salary greater than 70000.
7. Display employees whose salary is between 40,000 and 70,000.
8. Show employees hired after 2020-01-01
9. Show only the top 3 highest-paid employees.
10. Show employees in the Engineering department with a salary greater than 75,000.
11. Display employees who are currently active.
12. Display employee names in alphabetical order.
13. Display employees whose name starts with 'A'.
14. Display employees whose email ends with ".com".
15. Display employees sorted by salary in descending order.
16. Display employees sorted by hire date.
17. Display employees sorted first by department and then by salary.

Exercise 1.4: Write queries for the following (UPDATE and DELETE)

1. Alice Roy got a promotion — update her salary to 90000.
2. Change Gauri Iyer's department to Engineering .
3. Delete the record of David Nair (he left the company).
4. Mark all Marketing employees as inactive (is_active = FALSE)

Exercise 1.5: Create a table called *project* with the following columns

Column	Type	Constraint
project_id	INT	Primary Key, Auto Increment
project_name	VARCHAR(150)	Not Null
emp_id	INT	Foreign Key → employee(emp_id)
start_date	DATE	—
end_date	DATE	—
status	ENUM	'Active', 'Completed', 'On Hold'

Exercise 1.6: Insert 3 projects of your choice, assigning them to any existing employees.

1. Display all project details.
2. Display all completed projects.
3. Display project names and their status.
4. Display projects whose status is "On Hold".
5. Display projects whose end date is NULL (still ongoing).
6. Display projects that started in the year 2024.

7. Display projects sorted by start date.
8. Display projects sorted by status and then project name.

Exercise 1.7: Aggregate Functions

1. Find the total number of employees.
2. Find the average salary of employees.
3. Find the highest salary.
4. Find the lowest salary.
5. Find the total salary paid to all employees.
6. Count the number of active employees.
7. Count the number of projects.
8. Count the number of completed projects.
9. Find the average salary of employees in the IT department.
10. Find the highest salary among active employees.

Exercise 1.8: Join Queries

1. Display employee names along with their project names.
2. Display employee name, department, and project status.
3. Display all employees even if they have no project assigned.
4. Display all projects even if no employee is assigned.
5. Display employee name and project start date.
6. Display employee name and project end date. _____
7. Display employee name, project name, and project status.
8. Display names of employees working on completed projects.
9. Display names of employees working on active projects.
10. Display employees who are not assigned to any project.
11. Display projects that do not have an assigned employee.
12. Display employee names with the number of projects assigned.
13. Display employees who are assigned to more than one project.
14. Display employees who have never worked on a completed project.
15. Display the employee with the maximum number of projects.